

Tarang Singh

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An aspiring physicist with rigorous problem solving capabilities, and extensive research plus teaching experience

PROFESSIONAL EXPERIENCE

Tata Institute of Fundamental Research

Colaba, Mumbai

Winter Project, Dr. Mandar Deshmukh

October 2022 - November 2022

- Exfoliating WSe₂ and h-BN to study the 2-dimensional properties
- Extensive experimental research experience under professor Mandar Deshmukh by working in his condensed matter lab and clean room

National Initiative for Undergraduate Science 19.1, TIFR-HBCSE

Trombay, Mumbai

December 2022- January 2023

- Attending a series of lectures and experiments designed by scientist from various top research institutes in India designed to give undergraduate students a better exposure in research conducted in physical sciences
- Selected as one of 60 students from all over India to attend the initiative

National Initiative for Undergraduate Science 19.2, TIFR-HBCSE

Trombay, Mumbai

Research Project, Dr. Sudhir Jain

2023

- Selected through a test conducted for the 60 students selected through the initiative
- Understood the aspects of Quantum error corrections, basics of Quantum Computing and Genus Codes.

Tutor, Inkstall Educare Pvt Ltd.

Western Suburbs, Mumbai

June 2024-Ongoing

- Teaching VI to XII grade IGCSE, IBDP, ICSE and CBSE board and making them feel and understand the magic of math and physics

EDUCATION

Podar International School, ICSE

Kalyan, Thane

Primary and Secondary Schooling

2008-2019

B.K Birla College, HSC

Kalyan, Thane

Higher Secondary Schooling

2019-2021

D.G Ruparel College

Matunga, Mumbai

Bachelors in Science

2021-2024

Indian Institute of Technology (IIT-BHU)

Varanasi, Uttar Pradesh

Masters in Science

2025-2027

ACHIEVEMENTS

- Selected for ICTS-TIFR interview stage (**top 40-50** candidates in India through TIFR-GS written test)
- Qualified for TIFR written test (**top 123** in TIFR-GS, 2025)
- All India Rank **343**, JEST-2024
- All India Rank **437**, JAM-2025

- Class Representative, MSc (Batch of 2025), IIT-BHU
- Bronze Medal: Undergraduate Math Poster Competition, D.G. Ruparel College
- Bronze Medal: Undergraduate Intercollege Physics Quiz Competition, K.C. College, Mumbai
- Qualified Homi Bhabha Balvaidyanik Competition, Grade VI
- First Prize: Times Scholastic Aptitude Test

PARTICIPATIONS

- VIBGYOR 2023: Built a model demonstrating the effect of body's shape on motion in fluids (viscous forces) and constructed a rocket launcher capable of reaching the second floor of a standard building
- AVISHKAR 2022: Designed and built a Van de Graaff generator from inexpensive materials, achieving a potential difference of approximately 3 kilovolts
- Basic Astronomy Course: Completed at D.G. Ruparel College.
- Advanced Astronomy Course: Completed at D.G. Ruparel College
- Published Article: Featured in "Ganitananda", a mathematics journal by D.G. Ruparel College, with an article titled "Trigonometric Identities: and where did they come from?"
- IAPT Summer School 2022: Attended a summer school focused on problem-solving, organized by the Indian Association of Physics Teachers

PROFICIENCIES

- Lab techniques, CST, Python, MATLAB, LaTeX and extensive problem solving skills
- Understanding quantum mechanical systems for practical purposes using condensed matter physics